

# Technical Document #649: Database Systems - Comprehensive Overview

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NoSQL databases handle unstructured and semi-structured data. Document stores, key-value stores, and graph databases are common NoSQL types.

Database caching stores frequently accessed data in memory. Redis and Memcached are popular caching solutions.

Relational databases organize data into tables with rows and columns. SQL is used to query and manipulate structured data.

Database replication copies data across multiple servers. It improves availability and provides fault tolerance.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Remote work technologies have transformed the workplace. Collaboration tools and cloud services enable distributed teams to work effectively from anywhere.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

The rapid advancement of technology has transformed how organizations approach problem-solving and innovation. Companies must adapt to stay competitive in an ever-evolving landscape.

## Key Takeaways:

- NoSQL databases handle unstructured and semi-structured data.
- Sharding distributes data across multiple database servers.
- ACID properties (Atomicity, Consistency, Isolation, Durability) ensure reliable transaction processing in databases.